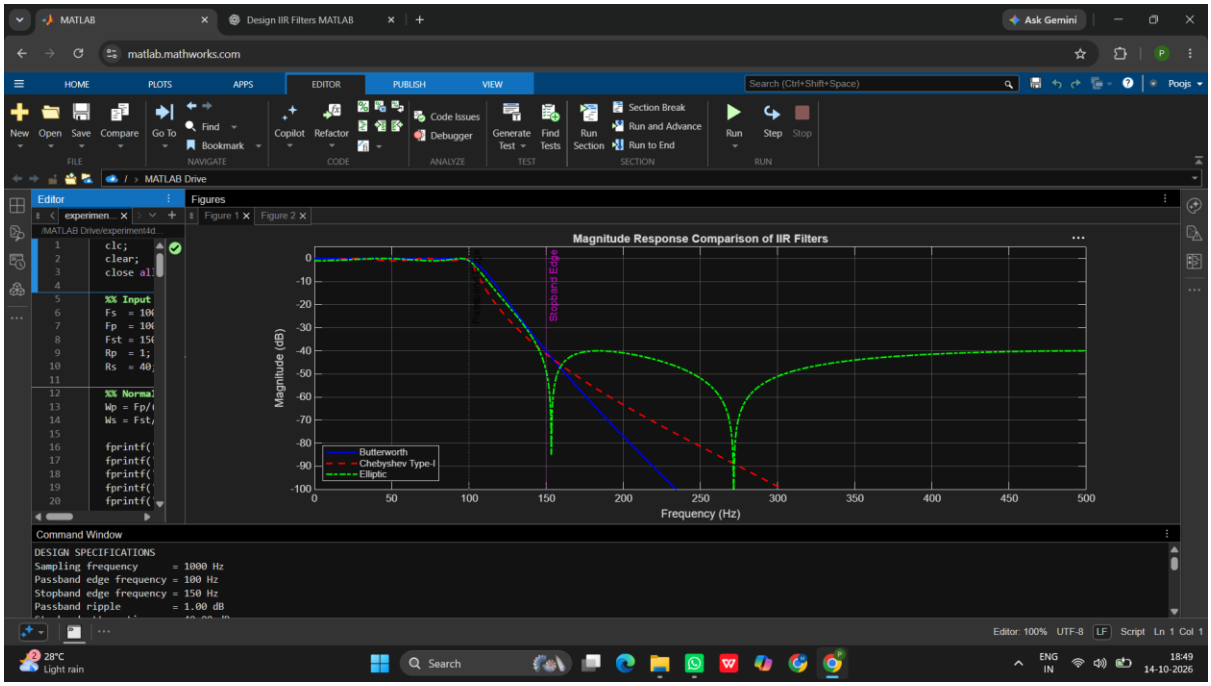
EXPERIMENT 4

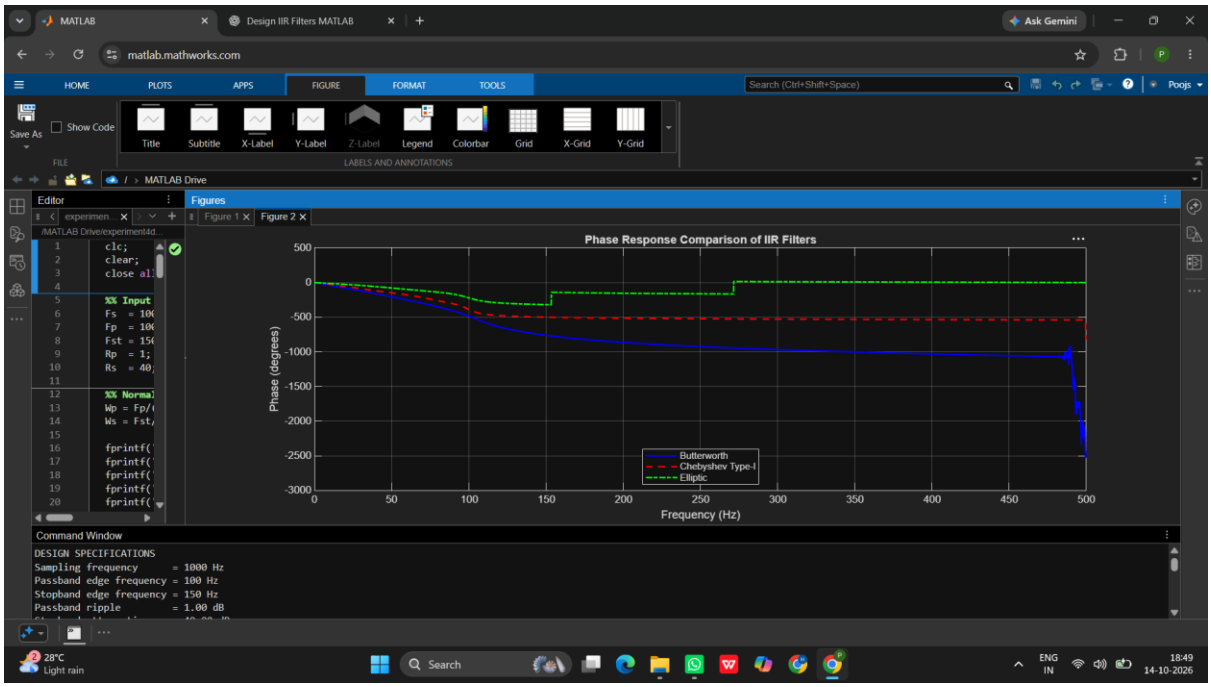
NAME: Pooja V

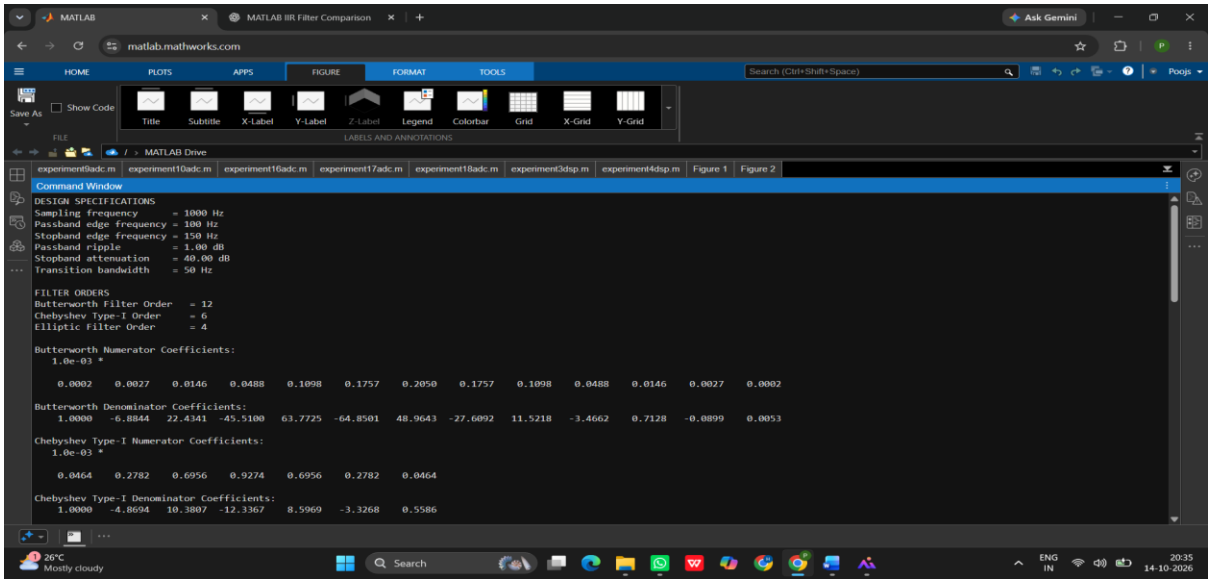
REG NO: 192512463

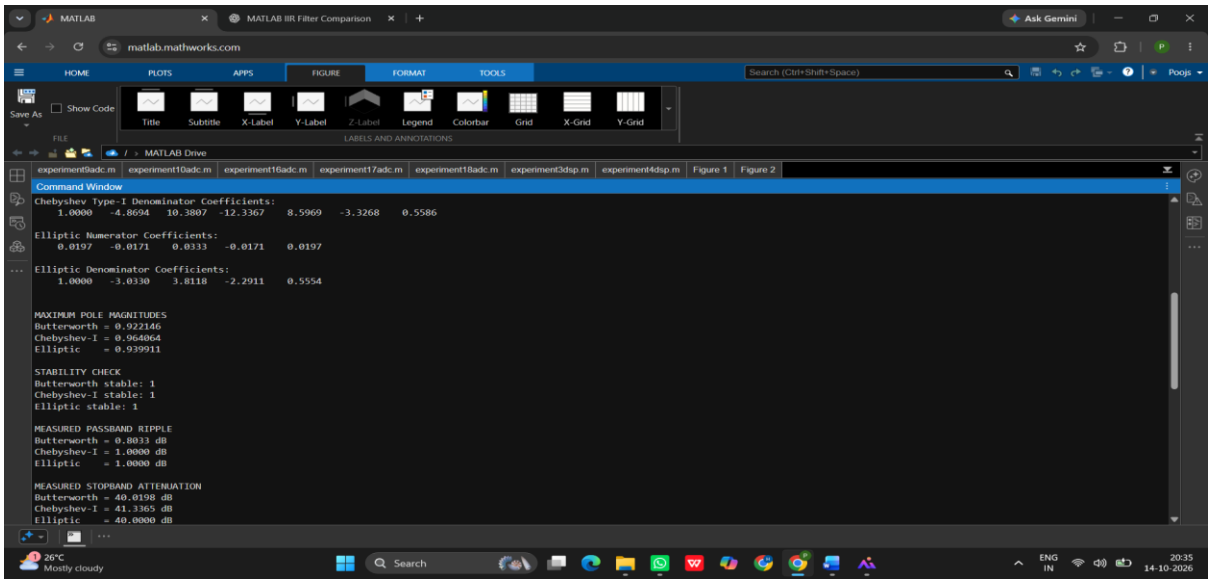
PROGRAM:

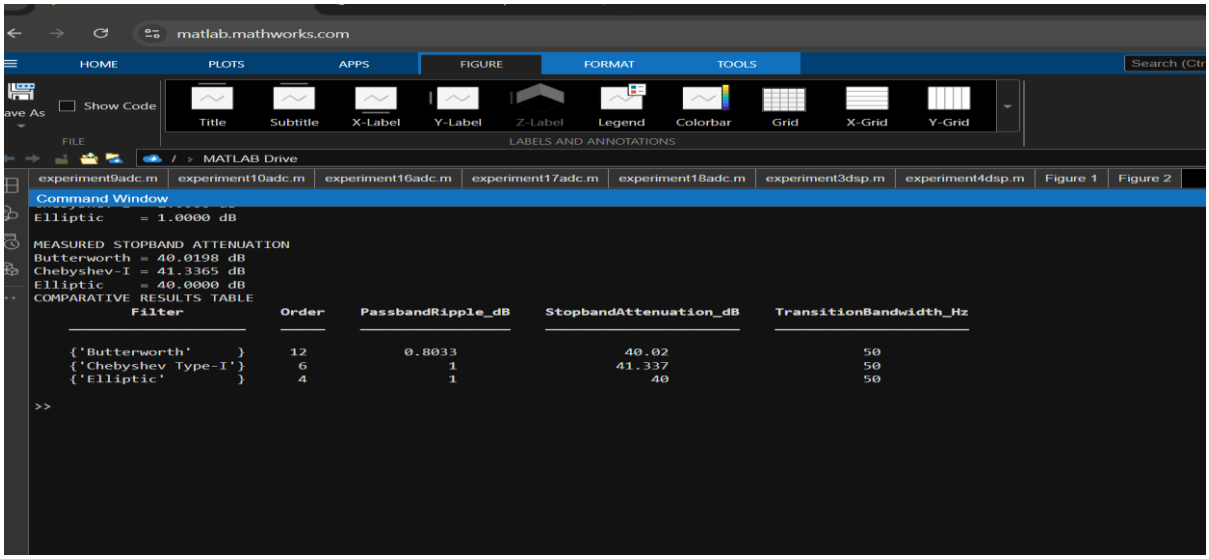
```
clear;
close all;
%% Input Specifications
Fs  = 1000;      % Sampling frequency in Hz
Fp  = 100;       % Passband edge frequency in Hz
Fst = 150;       % Stopband edge frequency in Hz
Rp  = 1;         % Passband ripple in dB
Rs  = 40;        % Stopband attenuation in dB
...
fprintf('\nMAXIMUM POLE MAGNITUDES\n');
fprintf('Butterworth = %.6f\n', max(abs(pB)));
fprintf('Chebyshev-I = %.6f\n', max(abs(pC)));
fprintf('Elliptic    = %.6f\n', max(abs(pE)));
%% Stability Check
fprintf('\nSTABILITY CHECK\n');
fprintf('Butterworth stable: %d\n', all(abs(pB) < 1));
```


Magnitude Response Comparison of IIR Filters
Stopband Edge
Magnitude (dB)
Frequency (Hz)
Butterworth
Chebyshev Type-I
Elliptic
DESIGN SPECIFICATIONS
Sampling frequency = 1000 Hz
Passband edge frequency = 100 Hz
Stopband edge frequency = 150 Hz
Passband ripple = 1.00 dB


Phase Response Comparison of IIR Filters
Phase (degrees)
Frequency (Hz)
Butterworth
Chebyshev Type-I
Elliptic
DESIGN SPECIFICATIONS
Sampling frequency = 1000 Hz
Passband edge frequency = 100 Hz
Stopband edge frequency = 150 Hz
Passband ripple = 1.00 dB


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experiment10adc.m
experiment16adc.m
experiment17adc.m
experiment18adc.m
experiment3dsp.m
experiment4dsp.m
Figure 1
Figure 2
Command Window
DESIGN SPECIFICATIONS
Sampling frequency = 1000 Hz
Passband edge frequency = 100 Hz
Stopband edge frequency = 150 Hz
Passband ripple = 1.00 dB
Stopband attenuation = 40.00 dB
Transition bandwidth = 50 Hz
FILTER ORDERS
Butterworth Filter Order = 12
Chebyshev Type-I Order = 6
Elliptic Filter Order = 4
Butterworth Numerator Coefficients:
1.0e-03 *
0.0002 0.0027 0.0146 0.0488 0.1098 0.1757 0.2050 0.1757 0.1098 0.0488 0.0146 0.0027 0.0002
Butterworth Denominator Coefficients:
1.0000 -6.8844 22.4341 -45.5100 63.7725 -64.8501 48.9643 -27.6092 11.5218 -3.4662 0.7128 -0.0899 0.0053
Chebyshev Type-I Numerator Coefficients:
1.0e-03 *
0.0464 0.2782 0.6956 0.9274 0.6956 0.2782 0.0464
Chebyshev Type-I Denominator Coefficients:
1.0000 -4.8694 10.3807 -12.3367 8.5969 -3.3268 0.5586
26°C
Mostly cloudy
Search
ENG
IN
20:35
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experiment4dsp.m
Figure 1
Figure 2
Command Window
Chebyshev Type-I Denominator Coefficients:
1.0000 -4.8694 10.3807 -12.3367 8.5969 -3.3268 0.5586
Elliptic Numerator Coefficients:
0.0197 -0.0171 0.0333 -0.0171 0.0197
Elliptic Denominator Coefficients:
1.0000 -3.0330 3.8118 -2.2911 0.5554
MAXIMUM POLE MAGNITUDES
Butterworth = 0.922146
Chebyshev-I = 0.964064
Elliptic = 0.939911
STABILITY CHECK
Butterworth stable: 1
Chebyshev-I stable: 1
Elliptic stable: 1
MEASURED PASSBAND RIPPLE
Butterworth = 0.8033 dB
Chebyshev-I = 1.0000 dB
Elliptic = 1.0000 dB
MEASURED STOPBAND ATTENUATION
Butterworth = 40.0198 dB
Chebyshev-I = 41.3365 dB
Elliptic = 40.0000 dB
26°C
Mostly cloudy
Search
ENG
IN
20:35
14-10-2026


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Figure 1
Figure 2
Command Window
Elliptic = 1.0000 dB
MEASURED STOPBAND ATTENUATION
Butterworth = 40.0198 dB
Chebyshev-I = 41.3365 dB
Elliptic = 40.0000 dB
COMPARATIVE RESULTS TABLE
Filter Order PassbandRipple_dB StopbandAttenuation_dB TransitionBandwidth_Hz
{'Butterworth' } 12 0.8033 40.02 50
{'Chebyshev Type-I'} 6 1 41.337 50
{'Elliptic' } 4 1 40 50
>>